



Outcome switching in cohort studies of interventions: meta-epidemiological study

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ABSTRACT

OBJECTIVES

To study the prevalence and characteristics of outcome switching, the completeness of outcome prespecification, and factors associated with outcome switching in observational cohort studies of interventions.

DESIGN

Longitudinal meta-epidemiological study.

SETTING

Registry records and journal publications.

PARTICIPANTS

Controlled cohort studies investigating the effects of interventions. Eligible studies were registered on ClinicalTrials.gov within one month of their start date (2014-16) and had published results in peer reviewed journals by 2024.

MAIN OUTCOMES MEASURES

Firstly, proportion of studies with outcome switching identified by comparing the prespecified outcomes in the registry and those reported in the journal publication of results. Discrepancies were categorised as omission (prespecified primary outcomes not reported), downgrading (prespecified primary outcomes reported as non-primary), upgrading (prespecified non-primary outcomes reported as primary), and introduction of new primary outcomes (not registered as an outcome). Secondly, proportion of studies with completely prespecified primary outcomes, defined as registry entries that include the measurement variable, analysis metric, method of aggregation (the statistic summarising the outcome within each study group), and time point.

RESULTS

Of 9965 registration records screened, 124 eligible studies with results published between 2015

and 2024 were included. Only 30 studies (24%) completely prespecified their primary outcomes. Outcome switching occurred in 60 (48%) studies, but only two provided an explanation. The most common types of switching were omission (n=32, 26%) and downgrading (n=32, 26%), followed by the introduction of new primary outcomes (n=25, 20%), and upgrading (n=2, 2%). Among 57 studies with outcome switching other than omission (ie, outcome results were reported), statistically significant results were favoured in 77% (44/57) by introducing or upgrading a new significant primary outcome or downgrading a non-significant one. No study characteristics were significantly associated with outcome switching in multivariable logistic regression.

CONCLUSIONS

Outcome switching and inadequate outcome prespecification were common in cohort studies of interventions. Most changes were unexplained and favoured statistically significant results, raising concerns about potential selective reporting and highlighting the need for improved transparency in outcome reporting.

STUDY REGISTRATION

Open Science Framework (<https://osf.io/xn5zt/>).

Introduction

Over a third of published systematic reviews of therapeutic interventions include non-randomised studies of interventions, most commonly observational cohort studies, whose results are frequently pooled with those from randomised controlled trials.¹⁻⁴ This practice suggests that observational data are often treated as an important complement to data from randomised controlled trials, and may represent the only source of evidence when randomised controlled trials are not available.

Like randomised controlled trials, cohort studies of interventions should be guided by a protocol that prespecifies the main study outcomes. Changing these outcomes in the final publication, known as outcome switching, can reflect legitimate refinement; however, switching based on the direction or statistical significance of the findings introduces selective outcome reporting bias.⁵ Transparency practices such as prospective registration document key changes as studies evolve, enabling readers to take this into account when interpreting the study results.⁶ Although registration of cohort studies of interventions is increasing and leading medical journals have long advocated it,⁷⁻¹⁰ uptake remains low (approximately 14% in a 2023 published sample).¹¹ Empirical data quantifying outcome switching in cohort studies of

WHAT IS ALREADY KNOWN ON THIS TOPIC

Cohort studies of interventions are often included in systematic reviews and used to guide clinical practice, particularly when randomised trials are not available

Outcome switching is a potential source of bias whereby prespecified primary outcomes differ from those reported in publications of results, but its prevalence is unclear for cohort studies

WHAT THIS STUDY ADDS

Among 124 cohort studies of interventions registered on ClinicalTrials.gov, 48% had at least one primary outcome changed, newly introduced, or omitted in the publication compared with the registry record

Three quarters of instances of outcome switching favoured statistically significant results, and only two were acknowledged in the publications

interventions are limited, leaving a gap that can inform broader adoption of transparency practices.

Our main objective was to estimate the prevalence of outcome switching in cohort studies of interventions and describe its characteristics by comparing primary outcomes prespecified in a public registry versus those subsequently reported in published articles. Other objectives were to assess the completeness of primary outcome prespecification and to identify study characteristics associated with outcome switching.

Methods

In this meta-epidemiological study, we identified cohort studies evaluating the effects of interventions (eg, drugs, procedures, counselling) on health related outcomes and compared the study outcomes defined in ClinicalTrials.gov and published articles. We focused on this international registry because it has the largest volume of observational studies among the eight registries that offer filters for observational studies.¹⁰ The study sample was created, and analytical methods were prespecified in a registered protocol (<https://osf.io/xn5zt/>) in 2024. Data extraction and analysis were conducted in 2025.

Study selection

We included controlled cohort studies of interventions that were prospectively registered on ClinicalTrials.gov and had a study start date between 1 January 2014 and 31 December 2016, as listed in the most recent registration record available at the time of data extraction. Eligible studies were required to have been registered within one month after the study start date or earlier; been marked as completed in the registry; and had their results published in a peer reviewed journal.

Using the registry's filters for "observational" studies and "completed recruitment," we identified an initial sample of registered observational studies. Two reviewers (ZS and CJ) then independently screened the title, study overview (including conditions being studied and a detailed study description), and study plan (including design details, study groups, interventions, and outcome measures) sections of each registry record to exclude studies that were not prospectively registered, lacked a control group or a health related outcome, or did not evaluate the effect of an intervention (eg, diagnostic or risk factor studies). Disagreements between reviewers were resolved by consensus or, when necessary, adjudicated by AH and A-WC.

We retrieved the corresponding publications through a stepwise search (figure S1), starting with the registration number. If no match was found, additional searches used combinations of intervention terms, health conditions, and the principal investigator's name.

Data collection

From the registration records, we classified the type of intervention as drug, procedure or device, or

behavioural (eg, counselling, education). We extracted all listed outcomes in the latest prospective version (latest version submitted within one month after the study start date), and the most recent version available at the time of data extraction, including their specification hierarchy (primary, secondary, other), measurement variable, measure description, and time point. We dichotomised the study design as a retrospective cohort (cohort constructed after the outcome has occurred) and prospective cohort (cohort constructed before outcome occurs). We categorised study funders as industry if any sponsor or collaborator was a pharmaceutical or device company.

For studies with several publications, we used an algorithm (figure S2) to identify the main publication. We excluded publications that did not report intervention effects or main results (eg, baseline, interim, or secondary analyses). We extracted the published outcome definitions and results with the same approach used for registered outcomes. We considered a reported outcome to be primary if it was explicitly described as "primary" or "main." If these terms were not found, we used the outcome mentioned for sample size calculation. When a paper reported only one outcome, we treated it as the primary outcome. When no primary outcomes were identified, all reported outcomes were classified as "unspecified."

We classified published study findings as statistically significant if the results for the primary outcome indicated that the experimental intervention was statistically superior to the control in superiority studies, not inferior in non-inferiority studies, or equivalent in equivalence studies based on the criteria defined by the authors. For studies without a clearly stated conceptual framework, statistical significance was based on the threshold defined by the authors. If no primary outcome was specified, any outcome that met the authors' definition of statistical significance was considered a significant finding.

Statistical analysis

The main objective of this study was to estimate the prevalence of outcome switching, measured as the proportion of studies that exhibited discrepancies in the primary outcome between the latest prospective registration record and published reports. We categorised the discrepancies as omission (registered primary outcome was not reported), downgrading (registered primary outcome was reported as secondary or unspecified), upgrading (registered secondary or unspecified outcome was reported as primary), and introduction of a new primary outcome that was not listed as an outcome in the registry.^{5 12} These categories are not mutually exclusive for a given study. An outcome was considered new if it differed from the registered primary outcome in terms of any of the four outcome components: measurement variable, analysis metric (eg, time to event, change from baseline), method of aggregation (ie, the statistic used to summarise the outcome data at group level such as mean or proportion), or time point.¹³ Outcome

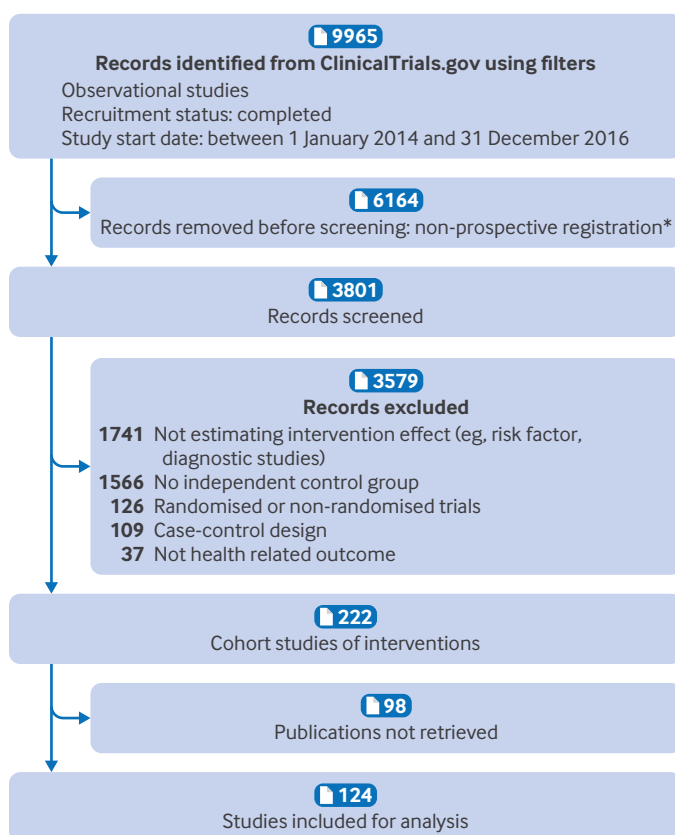


Fig 1 | Flowchart showing selection of prospectively registered cohort studies of interventions. *Registered more than one month after the study start date

switching was assessed independently by ZS and CJ, with disagreements resolved by consensus. To assess the impact of assumptions used to identify primary outcomes, we performed a sensitivity analysis limited to studies in which an outcome was explicitly labelled as primary in the publication.

We evaluated the completeness of registered primary outcome definitions at the study level. A study was deemed complete if at least one primary outcome specified all four components mentioned above.

To explore whether outcome switching favoured the statistical significance of the results, we calculated the proportion of studies in which a new statistically significant outcome was introduced or upgraded, or a non-significant one was downgraded. These switching patterns shift the weight of reported primary findings towards statistically significant results.¹⁴ Omitted outcomes were not included in this calculation because the results were not available.

Furthermore, we conducted multivariable logistic regression analyses to assess the association between outcome switching and six predefined study characteristics, including funder type (industry *v* non-industry), study design (prospective cohort *v* retrospective cohort), target sample size (continuous), statistical significance of published results (at least one significant *v* all non-significant), intervention type (drug or biological agent *v* other), and journal impact

factor in 2023 (continuous). A two sided P value <0.05 was considered statistically significant. All statistical analyses were performed using R version 4.5.0.

Sample size considerations

We estimated that 60 events (ie, studies with outcome switching) were required for the logistic regression analysis to achieve 10 events per covariate. We conservatively assumed a 50% prevalence of outcome switching that produces the widest possible confidence interval and is close to the reported median prevalence of 54% observed in clinical trials,¹⁵ and therefore targeted a sample of 120 registered studies with published results. Under this assumption, the sample size provides an upper bound of $\pm 9\%$ for the 95% confidence interval width for the estimated prevalence. We started with a convenience sample of studies with start dates in 2014-15 to allow sufficient time for publication, and expanded to 2016 to reach the target sample size.

Patient and public involvement

This meta-epidemiological study analysed methodological rather than clinical data (ie, outcome reporting practices). No patients or members of the public were involved in setting the research question, selecting outcome measures, developing plans, analysing or interpreting the findings, or writing up the results.

Results

Among the 9965 studies registered with a start date from 2014 to 2016, we identified 222 controlled, prospectively registered cohort studies of interventions (fig 1). Of these studies, 124 had results published in a peer reviewed journal (101 studies with publications matched based on registration number) between January 2015 and October 2024 and were included in our analyses.

Characteristics of included studies

Nearly 85% (n=104) of the studies were led by researchers in North America and Europe (table 1). Clinical content varied considerably, with the most frequent specialties being cardiology (n=23, 19%), obstetrics and gynaecology (n=13, 10%), and neurology (n=12, 9%). Amendments to registry records were common, with a median of seven versions per study (interquartile range 4-11). Two thirds of the studies received only non-industry funding. Most studies (n=89, 72%) adopted a prospective cohort design and half investigated drugs or biological agents. All studies provided a target sample size (median 300, interquartile range 100-1000). Only 27 (22%) studies made their full protocol publicly available as a published journal article (n=13, 10%) or an attachment in the registry (n=12, 10%), or both (n=2, 2%); none included the protocol as a supplement to the primary publication of results. Most studies published their findings as a single article (median 1,

Table 1 | Characteristics of included studies (n=124)

Characteristic	No of studies (%)*
Study start date	
2014	41 (33)
2015	39 (31)
2016	44 (35)
Region of corresponding author	
North America	58 (47)
Europe	46 (37)
Asia	18 (15)
South America	1 (0.8)
Oceania	1 (0.8)
Medical specialty†	
Cardiology	23 (19)
Obstetrics and Gynaecology	13 (10)
Neurology	12 (9)
Anaesthesiology	11 (9)
Oncology	9 (7)
Other	56 (46)
No of registry record versions, median (IQR)	7 (4-11)
Type of funder	
Industry (full or partial funding)	41 (33)
Only non-industry	83 (67)
Study design	
Prospective cohort‡	89 (72)
Retrospective cohort	35 (28)
Type of intervention	
Drug	62 (50)
Procedure or device	54 (44)
Behavioural	8 (7)
Target sample size in the registry, median (IQR)	300 (100-1000)
Publicly available full protocol	27 (22)
Uploaded in registry only	12 (10)
Published in journal only	13 (10)
In both registry and journal	2 (2)
As a supplement to main results paper	0 (0)
No of results papers published, median (IQR)	1 (1-2)
Impact factor of the journal where main paper published,§ median (IQR)	4 (3-6)
Type of journal where the main paper was published	
Specialty journal	110 (89)
General medical journal	14 (11)

IQR=interquartile range.

*Data are number (%) unless stated otherwise.

†Five most frequent specialties listed.

‡Including three studies with historical control group.

§As per journal citation reports in 2023.

interquartile range 1-2), primarily in specialty journals (n=110, 89%).

Completeness of primary outcome definition in registry

All the included studies listed at least one primary outcome in the registry. In the latest prospective version of registry entries, only 30 (24%) explicitly specified all four components of the primary outcome (measurement variable, analysis metric, method of aggregation, time point). All studies defined the time point at which the participants were assessed for the outcome measure. The outcome measurement variable (eg, blood pressure) was defined in 106 (85%) studies, whereas others provided only general concepts such as cardiac function or laboratory response. The analysis metric (eg, change from baseline) was specified in 75 (60%) studies. The least frequently defined element

was the method of aggregation (eg, mean, proportion), reported for the primary outcome of only 35 (28%) studies.

Primary outcomes were completely prespecified in a similar proportion of retrospective (26%, 9/35) versus prospective studies (24%, 21/89), and more often in drug studies (37%, 23/62) than non-pharmacological studies (11%, 7/62).

Prevalence and characteristics of outcome switching

In the main publication, primary outcomes were identified in 98 studies (79%)—explicitly labelled by the authors in 89 studies and inferred from the power calculation or a single outcome in nine studies. Statistically significant findings involving any type of outcome were reported in 98 (79%) studies.

Overall, 60 (48%) studies exhibited discrepancies between the registered and published primary outcomes (table 2). More than one type of outcome switching was found in 29 (23%) studies. Only two studies provided a rationale for the changes. The most common types of outcome switching were omission and downgrading, each occurring in 32 (26%) studies. For downgrading, at least one of the registered primary outcomes was later reported as an unspecified outcome in 24 studies (19%) or as a secondary outcome in eight studies (7%). In contrast, only two studies (2%) upgraded a registered secondary outcome to be a primary outcome in the paper. Aside from the omitted outcomes, all registered primary outcomes were reported with results sufficient for inclusion in a meta-analysis,¹² even though some were downgraded to non-primary.

Additionally, 25 (20%) studies introduced a total of 35 new primary outcomes that had not been registered as primary or secondary. These new outcomes differed from the registered ones mainly in the measurement variable (28/35, 80%), followed by the time point (21/35, 60%), analysis metric (4/35, 11%), and method of aggregation (2/35, 6%).

To explore whether outcome switching favoured statistically significant results, three studies involving only omission were excluded because of unknown results of the omitted outcomes. Among the remaining 57 studies, 44 (77%) favoured statistically significant results by downgrading a statistically non-significant primary outcome (54 of 72 downgraded outcomes across 23 studies), upgrading a statistically significant secondary outcome (two outcomes from two studies), or introducing a new statistically significant primary outcome (24 of 32 new outcomes from 22 studies). Numbers denoting studies are not mutually exclusive.

In the sensitivity analysis limited to the 89 studies in which the author explicitly labelled an outcome as primary in the publication, the prevalence of outcome switching dropped to 35% (n=31). This decrease is largely because of the removal of studies (n=17) in which outcome switching was classified as downgrading only because the publication did

Table 2 | Prevalence and characteristics of outcome switching when comparing registration records with published articles

Prevalence and characteristics	Definition	Examples in registry	Examples in publication	No of studies (%)
Any primary outcome switching (n=124 studies)*	—	—	—	60 (48)
Type of outcome switching (n=124 studies)				
Omission	At least one registered primary outcome not reported	Primary outcomes: treated non-tuberculous mycobacterium disease; hospitalised respiratory infection	Primary outcome: hospitalised respiratory infection Secondary outcome: all cause mortality	32 (26)
Downgrading	At least one registered primary outcome reported as non-primary outcome†	Primary outcomes: change in GMFM-66 scores; change in PEDI scores‡	Primary outcome: change in GMFM-66 scores Secondary outcome: change in PEDI scores	32 (26)
Upgrading	At least one registered non-primary outcome reported as primary outcome	Primary outcome: objective dry eye scores. Secondary outcome: subjective dry eye scores	Primary outcome: subjective and objective dry eye scores Secondary outcome: margin reflex distance	2 (2)
New primary outcome	New outcomes not listed in registry reported as primary	Primary outcome: change in glyocalyx thickness at 24 hours after surgery	Primary outcome: change in glyocalyx thickness at two hours after surgery	25 (20)
Component of outcome definitions that differed between published and registered primary outcomes (n=35 outcomes from 25 studies)§				
Measurement variable	Measurement collected directly from participants	Number of thromboembolic events and bleeding complications	Composite of stroke, systemic embolism, and death	28 (80)
Analysis metric	Format of outcome data that will be used from each participant for analysis	Change in leg volume from baseline	Incidence of lymphedema of lower limbs	4 (11)
Method of aggregation	Statistic that will be used to summarise outcome data at group level	Frequency of postpartum depression at two weeks postpartum (proportion)	Change in depression scores from baseline (mean)	2 (6)
Time point	Time point that a participant was analysed for measurement variable	During five months after vaccination	During risk intervals 0-2, 0-13, 0-41, and 0-180 days after vaccination	21 (60)

*More than one type of outcome switching found in 29 (23%) studies.

†At least one of registered primary outcomes later reported as unspecified outcomes in 24 studies (19%) or as secondary outcomes in eight studies (7%).

‡GMFM-66 (Gross Motor Function Measure) and PEDI (Paediatric Evaluation of Disability Inventory) are distinct measurement variables used to evaluate physical and functional status in paediatric populations.

§Some studies had differences in several components.

not label any outcome as primary, even though the prespecified primary outcomes were reported.

Factors associated with outcome switching

No statistically significant associations were found between outcome switching and various study characteristics (table 3).

Discussion

Principal findings

In 124 cohort studies of interventions registered in ClinicalTrials.gov, fewer than one quarter fully prespecified their primary outcomes in the registry. Sixty studies (48%) had discrepancies in primary outcomes between the publication and the registration record. Most instances of outcome switching favoured the reporting of statistically significant results.

Other studies

We found a higher prevalence of outcome switching (48%) than reported in previous reviews of observational studies, and our estimate falls within the range reported for randomised controlled trials (10-68%).¹⁶ Previous meta-research involving observational studies has been limited to specific specialty areas, outcome types, or high impact journals, and often included retrospectively registered studies. For example, Baudart and colleagues found 114 (85.7%) of 133 observational studies adequately reported the registered outcomes; however, their analysis focused exclusively on harms.¹⁷ Similarly, another review of cancer related observational

studies reported 81.9% (77/94) concordance between registered and published primary outcomes.¹⁸ This high concordance may be attributable to the common use of standardised outcomes in cancer studies.¹⁹

We considered any change in one of the four outcome components as a discrepancy. This broader approach, compared with previous work focusing on the outcome measure alone, captures more instances of outcome switching. We consider this appropriate because these components jointly define a distinct outcome.²⁰ Nonetheless, caution is warranted when comparing estimates across studies using different criteria.

Similar to our results, previous studies did not find significant associations between outcome switching and funding source, journal impact factor, study design, or sample size.²¹⁻²³ We removed a prespecified covariate “number of study centres (single vs. multiple)” from our analysis because some included studies did not operate through discrete study centres (eg, population level sampling).

Strengths

We adopted a comprehensive classification for outcome reporting discrepancies, adapted from previous similar assessments in randomised controlled trials.^{12 14 22 24} We also focused on a broad sample of hypothesis driven cohort studies evaluating healthcare interventions; although the low inclusion rate (1.2%, 124/9965) indicates these studies form a small subset of registered observational studies, they are particularly relevant to clinical journals and systematic reviews that inform practice.^{1 25}

Table 3 | Association between study characteristics and outcome switching

Study characteristic	Studies (n=117)*	Published with outcome switching†	Adjusted odds ratio‡ (95% CI)
Intervention type			
Drug or biological agent	57	27 (47)	0.84 (0.37 to 1.88)
Other	60	29 (48)	Reference
Target sample size,§ median (IQR)	300 (100-1000)	330 (100-950)	0.99 (0.82 to 1.16)
Design			
Retrospective cohort	33	19 (58)	1.82 (0.71 to 4.80)
Prospective cohort	84	37 (44)	Reference
Type of funder			
Industry involved	38	18 (47)	0.99 (0.43 to 2.29)
Non-industry	79	38 (48)	Reference
Journal impact factor , median (IQR)	4 (3-6)	4 (3-7)	0.99 (0.82 to 1.16)
Statistical significance of published results**			
At least one significant	97	46 (47)	0.94 (0.35 to 2.52)
All non-significant	20	10 (50)	Reference

Data are numbers (%) unless stated otherwise.

CI=confidence interval; IQR=interquartile range.

*Three studies without any hypothesis testing owing to insufficient sample size and four studies published in journals without impact factor excluded.

†Row percentage.

‡Adjusted odds ratios from multivariable logistic regression adjusting for listed study characteristics.

§Target sample size was log transformed (natural log) before inclusion in regression model to account for highly skewed and sparse distribution of data.

Odds ratios reflect change in odds per one unit increase in log transformed sample size.

||Per five point increase.

**Based on primary outcome; if no primary outcome specified, then based on any outcome.

Mechanisms and causality

The motivations behind outcome switching remain unclear. A taxonomy study summarised 12 determinants of selective outcome reporting, with a focus on positive findings, poor research design, and dependence upon sponsors as the most predominant.²⁶ In line with findings in the context of randomised controlled trials,^{12 14} most outcome changes in our sample reflected a preference for statistically significant results, suggesting potential reporting bias. Nevertheless, we were unable to confidently estimate the prevalence of this bias owing to unknown results of omitted outcomes. Outcomes may have been dropped or replaced because of unavailable data, statistical modelling constraints, or peer review feedback, which are scientifically sound. Therefore, we do not interpret our findings as a direct measure of selective outcome reporting bias but rather as empirical evidence of unacknowledged outcome switching.

Implications

If performed to highlight favourable results, outcome switching can introduce reporting bias that leads to research waste, a distorted evidence base, and misguided clinical decision making.⁵ For example, a study by Kirkham and colleagues found that among 81 systematic reviews with a single meta-analysis of randomised controlled trials, outcome reporting bias may have inflated treatment effect estimates by 20% or more in 19 reviews (23%).²⁷

Observational studies are often viewed as part of an evolutionary line of knowledge accrual that culminates in a clinical trial providing the definitive answer. In this context, outcome switching has received far less attention in observational studies than in randomised controlled trials, with concerns that registration could stifle their exploratory nature.^{10 28}

However, cohort studies of interventions evaluate specific interventions against comparators and test prespecified hypotheses, placing them closer to confirmatory than exploratory research²⁹; they provide the best available evidence when randomised controlled trials are not feasible and substantially shape the evidence base alongside randomised controlled trials.¹ Four high impact medical journal groups in 2024 published 63 systematic reviews of interventions that included observational studies, 33 of which relied exclusively on observational evidence. Of the remaining 30 reviews that included both observational studies and randomised controlled trials, 19 meta-analyses synthesised crude estimates from across designs.²⁵ Moreover, transparency practices document, rather than stifle, research creativity. Researchers can still refine outcomes and adapt analysis for data limitations. The additional work is to disclose and briefly justify deviations from the initial plan.⁶ For instance, when prespecified outcomes are modified, authors should explicitly report the change and rationale.

Given these considerations and the high prevalence of outcome switching, transparent outcome reporting is highly relevant to cohort studies of interventions. This aligns with current trends: since 2012, 19 of 25 published commentaries supported prospective registration of observational studies, seven registries now have specific policies about what information should be registered for these studies, and such registrations have increased rapidly.¹⁰ Although registration of observational studies is not currently mandated by medical journals, editorials in *The BMJ* and *The Lancet* have advocated for it.^{7 8}

Prospective registration deters selective reporting by enabling detection of deviations from prespecified study information in the reported results; however,

this relies on the assumption that outcomes are fully defined in the registration record.²⁰ Authors should therefore adhere to standards for outcome descriptions, such as the four outcome components recommended by the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) statement, to increase the level of specificity of registered outcome definitions.¹³ Although developed for randomised controlled trials, these standards are broadly applicable.

Additionally, protocols should be made publicly available. Beyond promoting transparency, access to protocols that clearly outline the predefined methods facilitates study replication, informed risk of bias assessment, and evaluation of the external applicability of study findings in practice.³⁰

Given the frequent outcome omission we observed, systematic reviewers should seek unpublished information from various sources such as registries, conference abstracts, author contact, and internet searches to identify unreported outcomes.³¹ Outcome reporting bias should be evaluated using risk-of-bias assessment tools.^{27 32} When reporting bias is suspected, sensitivity analyses can be conducted for key outcomes to assess the robustness of conclusions to outcome reporting bias.³³

Lastly, editors and reviewers should not only evaluate the methods prespecified in the protocol, but also compare them to the submitted reports of results. Authors should be required to explain any major discrepancies.

Challenges and limitations

In line with previous findings, protocols for the studies in our analysis were rarely made publicly available.³⁴ We had to rely on registration records to identify prespecified outcomes.

The true prevalence of outcome switching in our sample may be greater than what we observed owing to our inclusion of studies with vaguely predefined outcomes; this lack of specificity leaves greater room for undetectable outcome switching.³⁵ Among the 64 studies classified as having consistent outcome reporting, 10 (16%) may have undetected discrepancies because of inadequately prespecified outcomes. We retained these studies to generate a conservative estimate of outcome switching. Similar challenges were also noted in studies of outcome switching in trials.³⁶

Our sample is limited to a single registry and a narrow timeframe. Although registration of observational studies on ClinicalTrials.gov has increased markedly over the past two decades (from 314 in the year 1999 to 9775 in 2022), it remains optional and uncommon, with only 10% of observational studies published in top medical journals in 2023 being registered.³⁴ Because registered studies are likely to have better reporting than unregistered studies,²³ our findings likely underestimate the true extent of outcome switching across all cohort studies of interventions. Furthermore, some study protocols may have been revised without the updates being documented in the

registration record. Consequently, the prespecified outcomes we analysed may not represent the final study plan, potentially misclassifying pre-study modifications as outcome switching. We used the most recent prospective version of the registration record to mitigate this misclassification.

Future studies should include broader samples of cohort studies of interventions and more diverse sources of prespecified outcomes, particularly study protocols, to investigate outcome switching and its underlying mechanisms.

Conclusion

In conclusion, almost half of cohort studies of interventions showed outcome switching without a reported rationale, and most of the changes favoured statistically significant results. We also found that outcome definitions in the registry were frequently vague, and protocols were rarely made publicly available. These findings raise concerns about potential selective outcome reporting and underscore the need for improved transparency in cohort studies of interventions.

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Ethical approval: Institutional review board approval or informed consent was waived because the study was based on publicly available information and involved no patient records.

Data sharing: To avoid identifying individual studies, anonymised study level data and the data dictionary can be shared upon reasonable request to the corresponding author (zexing.song@mail.utoronto.ca) after publication.

Transparency: The corresponding author (the manuscript's guarantor) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant registered) have been explained.

Dissemination to participants and related patient and public communities: This study is a secondary analysis of publicly available information and did not involve direct participation of patients. Therefore, dissemination of study results to participants or patient/public communities was not planned. The article will be presented at scientific meetings and conferences, as well as other opportunities for dissemination.

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- 1 Page MJ, Shamseer L, Altman DG, et al. Epidemiology and reporting characteristics of systematic reviews of biomedical research: a cross-sectional study. *PLoS Med* 2016;13:e1002028. doi:10.1371/journal.pmed.1002028.
- 2 Shea BJ, Reeves BC, Wells G, et al. AMSTAR 2: a critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. *BMJ* 2017;358:j4008. doi:10.1136/bmj.j4008.
- 3 Axon E, Kanellopoulou A, Tsokani S, Livingstone N, Hilgart J, Richardson R. Incorporating non-randomized evidence in cochrane intervention reviews: a scoping review. *J Clin Epidemiol* 2025;183:111814. doi:10.1016/j.jclinepi.2025.111814.
- 4 Bun R-S, Scheer J, Guillo S, Tubach F, Dechartres A. Meta-analyses frequently pooled different study types together: a meta-epidemiological study. *J Clin Epidemiol* 2020;118:18-28. doi:10.1016/j.jclinepi.2019.10.013.
- 5 Thomas ET, Heneghan C. Catalogue of bias: selective outcome reporting bias. *BMJ Evid Based Med* 2022;27:370-2. doi:10.1136/bmjebm-2021-111845.
- 6 Naudet F, Patel CJ, DeVito NJ, et al. Improving the transparency and reliability of observational studies through registration. *BMJ* 2024;384:e076123. doi:10.1136/bmj-2023-076123.
- 7 Loder E, Groves T, Macauley D. Registration of observational studies. *BMJ* 2010;340:c950. doi:10.1136/bmj.c950.
- 8 The Lancet. Should protocols for observational research be registered? *Lancet* 2010;375:348. doi:10.1016/S0140-6736(10)60148-1.
- 9 Berlin JA, Fihn SD. Encouraging the registration of observational studies. *JAMA Netw Open* 2025;8:e2524181. doi:10.1001/jamanetworkopen.2025.24181.
- 10 Malmström D, Norlén S, Jespersen C, et al. Scoping review of registration of observational studies finds inadequate registration policies, increased registration, and a debate converging toward preregistration. *J Clin Epidemiol* 2025;180:111686. doi:10.1016/j.jclinepi.2025.111686.
- 11 Jespersen C, Song Z, Hróbjartsson A, et al. Registration of Observational Studies of Interventions: Prevalence, Characteristics, and Journal Policies. 2025. <https://peerreviewcongress.org/abstract/registration-of-observational-studies-of-interventions-prevalence-characteristics-and-journal-policies> (accessed 6 Mar 2026)
- 12 Chan A-W, Hróbjartsson A, Haahr MT, Gøtzsche PC, Altman DG. Empirical evidence for selective reporting of outcomes in randomized trials: comparison of protocols to published articles. *JAMA* 2004;291:2457-65. doi:10.1001/jama.291.20.2457.
- 13 Chan A-W, Boutron I, Hopewell S, et al. SPIRIT 2025 statement: updated guideline for protocols of randomized trials. *Nat Med* 2025;31:1784-92. doi:10.1038/s41591-025-03668-w.
- 14 Mathieu S, Boutron I, Moher D, Altman DG, Ravaud P. Comparison of registered and published primary outcomes in randomized controlled trials. *JAMA* 2009;302:977-84. doi:10.1001/jama.2009.1242.
- 15 Li G, Abbade LPF, Nwosu I, et al. A systematic review of comparisons between protocols or registrations and full reports in primary biomedical research. *BMC Med Res Methodol* 2018;18:9. doi:10.1186/s12874-017-0465-7.
- 16 TARG Meta-Research Group & Collaborators. Estimating the prevalence of discrepancies between study registrations and publications: a systematic review and meta-analyses. *BMJ Open* 2023;13:e076264. doi:10.1136/bmjopen-2023-076264.
- 17 Baudart M, Ravaud P, Baron G, Dechartres A, Haneef R, Boutron I. Public availability of results of observational studies evaluating an intervention registered at ClinicalTrials.gov. *BMC Med* 2016;14:7. doi:10.1186/s12916-016-0551-4.
- 18 Boccia S, Rothman KJ, Panic N, et al. Registration practices for observational studies on ClinicalTrials.gov indicated low adherence. *J Clin Epidemiol* 2016;70:176-82. doi:10.1016/j.jclinepi.2015.09.009.
- 19 Wilson MK, Karakasis K, Oza AM. Outcomes and endpoints in trials of cancer treatment: the past, present, and future. *Lancet Oncol* 2015;16:e32-42. doi:10.1016/S1470-2045(14)70375-4.
- 20 Butcher NJ, Monsour A, Mew EJ, et al. Guidelines for reporting outcomes in trial protocols: the SPIRIT-Outcomes 2022 Extension. *JAMA* 2022;328:2345-56. doi:10.1001/jama.2022.21243.
- 21 Holst M, Haslberger M, Yerunkar S, Strech D, Hemkens LG, Carlisle BG. Frequency of multiple changes to prespecified primary outcomes of clinical trials completed between 2009 and 2017 in German university medical centers: A meta-research study. *PLoS Med* 2023;20:e1004306. doi:10.1371/journal.pmed.1004306.
- 22 Jones CW, Keil LG, Holland WC, Caughey MC, Platts-Mills TF. Comparison of registered and published outcomes in randomized controlled trials: a systematic review. *BMC Med* 2015;13:282. doi:10.1186/s12916-015-0520-3.
- 23 Chan A-W, Pello A, Kitchen J, et al. Association of trial registration with reporting of primary outcomes in protocols and publications. *JAMA* 2017;318:1709-11. doi:10.1001/jama.2017.13001.
- 24 Hannink G, Gooszen HG, Rovers MM. Comparison of registered and published primary outcomes in randomized clinical trials of surgical interventions. *Ann Surg* 2013;257:818-23. doi:10.1097/SLA.0b013e3182864fa3.
- 25 Paul M, Olchowski J, Leibovici L. Systematic reviews of observational studies frequently conclude based on meta-analyses of biased results: standards must be improved. *J Clin Epidemiol* 2025;184:111840. doi:10.1016/j.jclinepi.2025.111840.
- 26 van der Steen JT, van den Bogert CA, van Soest-Poortvliet MC, et al. Determinants of selective reporting: a taxonomy based on content analysis of a random selection of the literature. *PLoS One* 2018;13:e0188247. doi:10.1371/journal.pone.0188247.
- 27 Kirkham JJ, Dwan KM, Altman DG, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. *BMJ* 2010;340:c365. doi:10.1136/bmj.c365.
- 28 Lash TL, Vandenbroucke JP. Should preregistration of epidemiologic study protocols become compulsory? Reflections and a counterproposal. *Epidemiology* 2012;23:184-8. doi:10.1097/EDE.0b013e318245c05b.
- 29 Fife DA, Rodgers JL. Understanding the exploratory/confirmatory data analysis continuum: Moving beyond the "replication crisis". *Am Psychol* 2022;77:453-66. doi:10.1037/amp0000886.
- 30 Chan A-W, Hróbjartsson A. Promoting public access to clinical trial protocols: challenges and recommendations. *Trials* 2018;19:116. doi:10.1186/s13063-018-2510-1.
- 31 Chan A-W. Out of sight but not out of mind: how to search for unpublished clinical trial evidence. *BMJ* 2012;344:d8013. doi:10.1136/bmj.d8013.
- 32 Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *BMJ* 2016;355:i4919. doi:10.1136/bmj.i4919.
- 33 Kirkham JJ, Altman DG, Chan A-W, Gamble C, Dwan KM, Williamson PR. Outcome reporting bias in trials: a methodological approach for assessment and adjustment in systematic reviews. *BMJ* 2018;362:k3802. doi:10.1136/bmj.k3802.
- 34 Leducq S, Zaki F, Hollestein LM, et al. The majority of observational studies in leading peer-reviewed medicine journals are not registered and do not have a publicly accessible protocol: a scoping review. *J Clin Epidemiol* 2024;170:111341. doi:10.1016/j.jclinepi.2024.111341.
- 35 Kahan BC, Jairath V. Outcome pre-specification requires sufficient detail to guard against outcome switching in clinical trials: a case study. *Trials* 2018;19:265. doi:10.1186/s13063-018-2654-z.
- 36 Zarín DA, Tse T, Williams RJ, Rajakannan T. Update on Trial Registration 11 Years after the ICMJE Policy Was Established. *N Engl J Med* 2017;376:383-91. doi:10.1056/NEJMs1601330.

Web appendix: Supplemental figures